

AKASHDEEP SINGH JIDA

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EDUCATION

Bachelor of Science, Electrical Engineering

GPA: 3.25/4.00

California State University, Sacramento, CA

Graduated May 2020

Associates in Mathematics

Fresno City College, Fresno, CA

Aug 2015 – Dec 2017

TECHNICAL SKILLS

- | | | | |
|----------------|--------------------|-----------------|----------------------|
| • Python | • Cadance Virtuoso | • AutoCAD | • Function Generator |
| • Verilog/VHDL | • Xilinx | • Quartus | • LaTeX |
| • C/C++ | • ModelSim | • Visual Studio | • FPGA |
| • Matlab | • PSpice | • PCB designing | • Raspberry Pi |
| • Arduino UNO | • STM32 | • XCAL | |

RELEVANT COURSEWORK

- | | |
|-------------------------|----------------------------|
| • Advanced Logic Design | • Computer Hardware Design |
| • CMOS and VLSI | • Power Control Drivers |
| • Signals and Systems | • Applied Electromagnetics |
| • Network Analysis | • Electronics |

WORK EXPERIENCE

California State University, Sacramento

Research Assistant

June 2020 – Present

- Assistant in literature review, technical writing and research on safety of mmWave communication.
- Ongoing work includes research on safeguarding 5G communication and publishing research findings.

Fresno City College, Fresno

Instructional Student Assistant – Tutor

Aug 2017 – Dec 2017

- Provided appropriate tutorials to meet each student's need.
- Marked and provided feedbacks to the students.

Telnet

RF Engineer (Volunteer)

June 2018 – July 2018

- Volunteered for contributing in the RF Design of the regional cellular network.
- Creating search sectors within their geographic area and working with the team to understand issue within the sector.

PROJECTS

Hydroponics System (*Senior Design Project*)

Aug 2020 - May 2020

- Designed, developed, tested, debugged and managed an autonomous hydroponics system.
- Arduino and Raspberry were used as microcontrollers to execute functions and remotely transfer database.

Cadence Virtuoso Layout

Sept 2019 - Dec 2019

- Designed, simulated and prepared layout of Inverter, Flip-Flop, 2 input NAND and Mirror adder.
- Verified its efficiency and functionality using waveforms, LVS and DRC reports.

Op-Amp Design Using PSPICE

Oct 2018 - Nov 2020

- A design-oriented trade-off was used to modify the operational amplifier to meet the desired results.
- Using a Telescopic amplifier to meet the requirement for specific voltage gain, CMRR and Unity Gain Bandwidth.
- The technology library of 0.13um for CMOS was used.

Arduino Ultrasonic Sensor Radar

Nov 2018 - Dec 2018

- Designed Ultrasonic Sensor radar using Arduino UNO, HC-SR04 sensor, servo on breadboard with coding in C++

Simulation of Verilog design using FPGA

Jan 2018 - May 2018

- Simulated Verilog and Test bench in Modelsim to get waveform and debug circuits that are implemented by FPGA.
- Designed 8-bit Carry Adder, BCD and Arithmetic Logic design.

PUBLICATION

Talwar R., Amala N., Medina G., Jida A.S., and Eltayeb M.E. (2020). *Exploiting Multi-Path for Safeguarding mmWave Communications Against Randomly Located Eavesdroppers*. ICSNC 2020 : The Fifteenth International Conference on Systems and Networks Communications

Link : (https://www.thinkmind.org/articles/icsnc_2020_2_80_28008.pdf)